

Assignment_4: k-Means for Clustering

Robert Andoh

2026-06-14

Introduction

This assignment applies k-Means clustering to financial data from 21 pharmaceutical firms. The purpose is to identify groups of firms with similar financial characteristics using numerical financial variables only.

The clustering model uses variables 1 to 9 from the dataset:

- Market capitalization
- Beta
- Price/earnings ratio
- Return on equity
- Return on assets
- Asset turnover
- Leverage
- Estimated revenue growth
- Net profit margin

The non-clustering variables, including median recommendation, headquarters location, and stock exchange, are used later to interpret the cluster patterns.

Load Required Packages

```
library(cluster)
library(factoextra)
```

```
## Loading required package: ggplot2
```

```
## Welcome to factoextra!
```

```
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practice
```

Import and Inspect Dataset

```
pharma <- read.csv("Pharmaceuticals.csv")
```

```
head(pharma)
```

```
##   Symbol      Name Market_Cap Beta PE_Ratio ROE ROA Asset_Turnover
## 1  ABT Abbott Laboratories    68.44 0.32    24.7 26.4 11.8      0.7
## 2  AGN   Allergan, Inc.     7.58 0.41    82.5 12.9  5.5      0.9
## 3  AHM   Amersham plc      6.30 0.46    20.7 14.9  7.8      0.9
## 4  AZN   AstraZeneca PLC    67.63 0.52    21.5 27.4 15.4      0.9
## 5  AVE   Aventis          47.16 0.32    20.1 21.8  7.5      0.6
## 6  BAY   Bayer AG         16.90 1.11    27.9  3.9  1.4      0.6
##   Leverage Rev_Growth Net_Profit_Margin Median_Recommendation Location Exchange
## 1    0.42     7.54         16.1      Moderate Buy      US      NYSE
## 2    0.60     9.16         5.5      Moderate Buy    CANADA    NYSE
## 3    0.27     7.05        11.2      Strong Buy      UK      NYSE
## 4    0.00    15.00        18.0      Moderate Sell    UK      NYSE
## 5    0.34    26.81        12.9      Moderate Buy    FRANCE    NYSE
## 6    0.00    -3.17         2.6      Hold      GERMANY    NYSE
```

```
dim(pharma)
```

```
## [1] 21 14
```

```
str(pharma)
```

```
## 'data.frame':    21 obs. of  14 variables:
## $ Symbol      : chr  "ABT" "AGN" "AHM" "AZN" ...
## $ Name        : chr  "Abbott Laboratories" "Allergan, Inc." "Amersham plc" "AstraZeneca PL
## $ Market_Cap  : num  68.44 7.58 6.3 67.63 47.16 ...
## $ Beta        : num  0.32 0.41 0.46 0.52 0.32 1.11 0.5 0.85 1.08 0.18 ...
## $ PE_Ratio    : num  24.7 82.5 20.7 21.5 20.1 27.9 13.9 26 3.6 27.9 ...
## $ ROE         : num  26.4 12.9 14.9 27.4 21.8 3.9 34.8 24.1 15.1 31 ...
## $ ROA         : num  11.8 5.5 7.8 15.4 7.5 1.4 15.1 4.3 5.1 13.5 ...
## $ Asset_Turnover : num  0.7 0.9 0.9 0.9 0.6 0.6 0.9 0.6 0.3 0.6 ...
## $ Leverage    : num  0.42 0.6 0.27 0 0.34 0 0.57 3.51 1.07 0.53 ...
## $ Rev_Growth   : num  7.54 9.16 7.05 15 26.81 ...
## $ Net_Profit_Margin : num  16.1 5.5 11.2 18 12.9 2.6 20.6 7.5 13.3 23.4 ...
## $ Median_Recommendation: chr  "Moderate Buy" "Moderate Buy" "Strong Buy" "Moderate Sell" ...
## $ Location     : chr  "US" "CANADA" "UK" "UK" ...
## $ Exchange     : chr  "NYSE" "NYSE" "NYSE" "NYSE" ...
```

Select Numerical Variables for Clustering

```
pharma_numeric <- pharma[, c(
  "Market_Cap",
  "Beta",
  "PE_Ratio",
  "ROE",
```

```

"ROA",
"Asset_Turnover",
"Leverage",
"Rev_Growth",
"Net_Profit_Margin"
)]

```

```
head(pharma_numeric)
```

```

##   Market_Cap Beta PE_Ratio  ROE  ROA Asset_Turnover Leverage Rev_Growth
## 1    68.44 0.32    24.7 26.4 11.8          0.7    0.42    7.54
## 2     7.58 0.41    82.5 12.9  5.5          0.9    0.60    9.16
## 3     6.30 0.46    20.7 14.9  7.8          0.9    0.27    7.05
## 4    67.63 0.52    21.5 27.4 15.4          0.9    0.00   15.00
## 5    47.16 0.32    20.1 21.8  7.5          0.6    0.34   26.81
## 6    16.90 1.11    27.9  3.9  1.4          0.6    0.00   -3.17
##   Net_Profit_Margin
## 1             16.1
## 2              5.5
## 3             11.2
## 4             18.0
## 5             12.9
## 6              2.6

```

```
summary(pharma_numeric)
```

```

##   Market_Cap      Beta      PE_Ratio      ROE
## Min.   : 0.41   Min.   :0.1800   Min.   : 3.60   Min.   : 3.9
## 1st Qu.: 6.30   1st Qu.:0.3500   1st Qu.:18.90   1st Qu.:14.9
## Median :48.19   Median :0.4600   Median :21.50   Median :22.6
## Mean   :57.65   Mean   :0.5257   Mean   :25.46   Mean   :25.8
## 3rd Qu.:73.84   3rd Qu.:0.6500   3rd Qu.:27.90   3rd Qu.:31.0
## Max.   :199.47   Max.   :1.1100   Max.   :82.50   Max.   :62.9
##   ROA      Asset_Turnover      Leverage      Rev_Growth
## Min.   : 1.40   Min.   :0.3    Min.   :0.0000   Min.   : -3.17
## 1st Qu.: 5.70   1st Qu.:0.6    1st Qu.:0.1600   1st Qu.:  6.38
## Median :11.20   Median :0.6    Median :0.3400   Median :  9.37
## Mean   :10.51   Mean   :0.7    Mean   :0.5857   Mean   :13.37
## 3rd Qu.:15.00   3rd Qu.:0.9    3rd Qu.:0.6000   3rd Qu.:21.87
## Max.   :20.30   Max.   :1.1    Max.   :3.5100   Max.   :34.21
##   Net_Profit_Margin
## Min.   : 2.6
## 1st Qu.:11.2
## Median :16.1
## Mean   :15.7
## 3rd Qu.:21.1
## Max.   :25.5

```

Standardize the Numerical Variables

```
pharma_scaled <- scale(pharma_numeric)
```

```
head(pharma_scaled)
```

```
##      Market_Cap      Beta      PE_Ratio      ROE      ROA      Asset_Turnover
## [1,]  0.1840960 -0.80125356 -0.04671323  0.04009035  0.2416121      0.0000000
## [2,] -0.8544181 -0.45070513  3.49706911 -0.85483986 -0.9422871      0.9225312
## [3,] -0.8762600 -0.25595600 -0.29195768 -0.72225761 -0.5100700      0.9225312
## [4,]  0.1702742 -0.02225704 -0.24290879  0.10638147  0.9181259      0.9225312
## [5,] -0.1790256 -0.80125356 -0.32874435 -0.26484883 -0.5664461     -0.4612656
## [6,] -0.6953818  2.27578267  0.14948233 -1.45146000 -1.7127612     -0.4612656
##      Leverage Rev_Growth Net_Profit_Margin
## [1,] -0.2120979 -0.5277675      0.06168225
## [2,]  0.0182843 -0.3811391     -1.55366706
## [3,] -0.4040831 -0.5721181     -0.68503583
## [4,] -0.7496565  0.1474473      0.35122600
## [5,] -0.3144900  1.2163867     -0.42597037
## [6,] -0.7496565 -1.4971443     -1.99560225
```

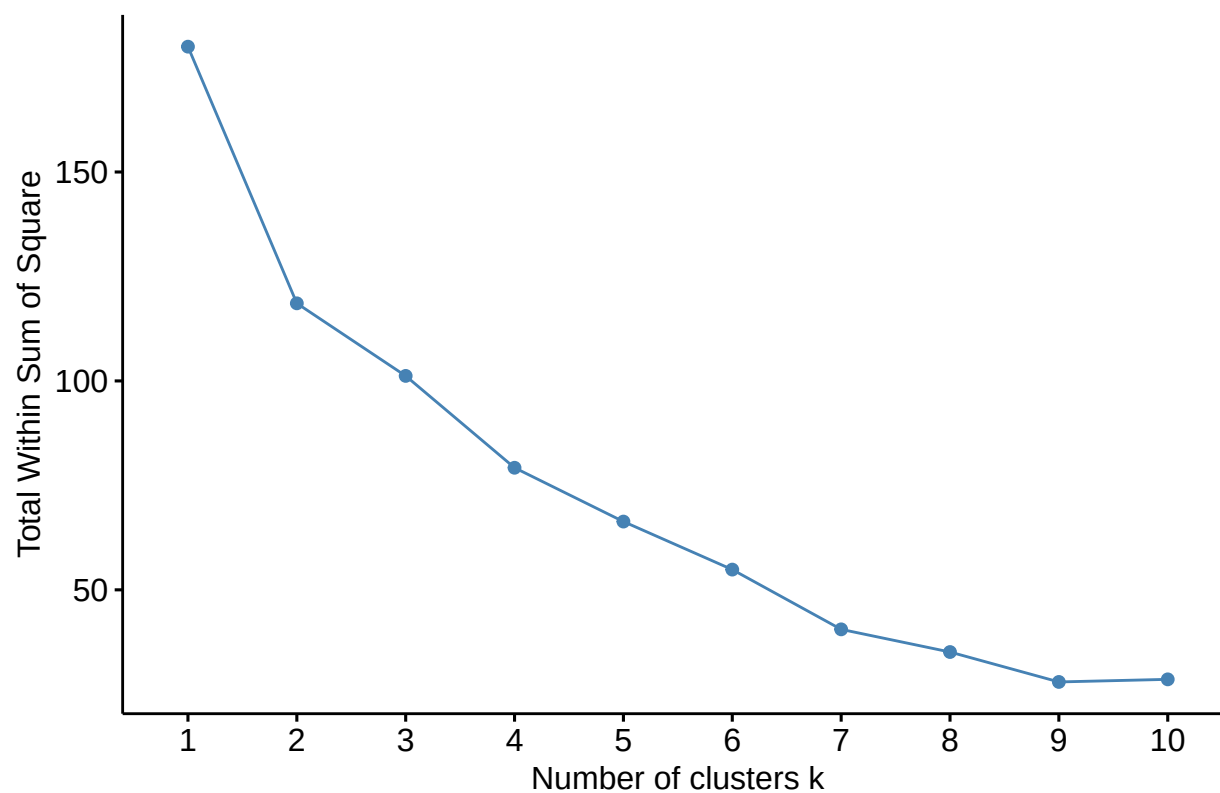
```
summary(pharma_scaled)
```

```
##      Market_Cap      Beta      PE_Ratio      ROE
## Min.      :-0.9768   Min.      :-1.3466   Min.      :-1.3404   Min.      :-1.4515
## 1st Qu.: -0.8763   1st Qu.: -0.6844   1st Qu.: -0.4023   1st Qu.: -0.7223
## Median : -0.1614   Median : -0.2560   Median : -0.2429   Median : -0.2118
## Mean    :  0.0000   Mean    :  0.0000   Mean    :  0.0000   Mean    :  0.0000
## 3rd Qu.:  0.2762   3rd Qu.:  0.4841   3rd Qu.:  0.1495   3rd Qu.:  0.3450
## Max.    :  2.4200   Max.    :  2.2758   Max.    :  3.4971   Max.    :  2.4597
##      ROA      Asset_Turnover      Leverage      Rev_Growth
## Min.      :-1.7128   Min.      :-1.8451   Min.      :-0.74966   Min.      :-1.4971
## 1st Qu.: -0.9047   1st Qu.: -0.4613   1st Qu.: -0.54487   1st Qu.: -0.6328
## Median :  0.1289   Median : -0.4613   Median : -0.31449   Median : -0.3621
## Mean    :  0.0000   Mean    :  0.0000   Mean    :  0.00000   Mean    :  0.0000
## 3rd Qu.:  0.8430   3rd Qu.:  0.9225   3rd Qu.:  0.01828   3rd Qu.:  0.7693
## Max.    :  1.8389   Max.    :  1.8451   Max.    :  3.74280   Max.    :  1.8862
## Net_Profit_Margin
## Min.      :-1.99560
## 1st Qu.: -0.68504
## Median :  0.06168
## Mean    :  0.00000
## 3rd Qu.:  0.82364
## Max.    :  1.49416
```

```
dim(pharma_scaled) # Determine the Optimal Number of Clusters
```

```
fviz_nbclust(
  pharma_scaled,
  kmeans,
  method = "wss"
) +
  ggtitle("Elbow Method for Selecting Number of Clusters")
```

Elbow Method for Selecting Number of Clusters



Build the k-Means Clustering Model

```
set.seed(123)

kmeans_model <- kmeans(
  pharma_scaled,
  centers = 3,
  nstart = 25
)

pharma$Cluster <- as.factor(kmeans_model$cluster)

table(pharma$Cluster)
```

```
##
##  1  2  3
##  4 11  6
```

```
pharma$Cluster <- as.factor(kmeans_model$cluster)

head(pharma)
```

```
##   Symbol      Name Market_Cap Beta PE_Ratio ROE  ROA Asset_Turnover
## 1   ABT Abbott Laboratories   68.44 0.32   24.7 26.4 11.8         0.7
## 2   AGN   Allergan, Inc.    7.58 0.41   82.5 12.9  5.5         0.9
```

```
## 3    AHM      Amersham plc      6.30 0.46      20.7 14.9  7.8      0.9
## 4    AZN      AstraZeneca PLC   67.63 0.52      21.5 27.4 15.4      0.9
## 5    AVE      Aventis          47.16 0.32      20.1 21.8  7.5      0.6
## 6    BAY      Bayer AG         16.90 1.11      27.9  3.9  1.4      0.6
##      Leverage Rev_Growth Net_Profit_Margin Median_Recommendation Location Exchange
## 1      0.42      7.54          16.1      Moderate Buy      US      NYSE
## 2      0.60      9.16           5.5      Moderate Buy     CANADA  NYSE
## 3      0.27      7.05          11.2      Strong Buy      UK      NYSE
## 4      0.00     15.00          18.0      Moderate Sell     UK      NYSE
## 5      0.34     26.81          12.9      Moderate Buy     FRANCE  NYSE
## 6      0.00     -3.17           2.6              Hold  GERMANY  NYSE
##      Cluster
## 1          2
## 2          1
## 3          1
## 4          2
## 5          3
## 6          1
```

```
cluster_membership <- as.data.frame(table(pharma$Cluster))
names(cluster_membership) <- c("Cluster", "Number_of_Firms")

cluster_membership
```

```
##      Cluster Number_of_Firms
## 1          1             4
## 2          2            11
## 3          3             6
```

Cluster Membership

The k-Means algorithm assigned the 21 pharmaceutical firms into three clusters.

Cluster	Number of Firms
Cluster 1	4
Cluster 2	11
Cluster 3	6

Cluster 2 contains the largest number of firms and represents the dominant financial profile in the dataset. Clusters 1 and 3 contain fewer firms and represent more specialized financial characteristics.

```
cluster_summary <- aggregate(
  pharma_numeric,
  by = list(Cluster = pharma$Cluster),
  mean
)

cluster_summary
```

##	Cluster	Market_Cap	Beta	PE_Ratio	ROE	ROA	Asset_Turnover	Leverage
## 1	1	21.75500	0.5950000	46.90000	11.3	5.100000	0.7500000	0.3050000
## 2	2	97.11364	0.4336364	20.95455	35.7	14.954545	0.8000000	0.3254545
## 3	3	9.23500	0.6483333	19.43333	17.3	5.983333	0.4833333	1.2500000

##	Rev_Growth	Net_Profit_Margin
## 1	7.01000	6.65000
## 2	10.16455	20.17273
## 3	23.49000	13.51667

Cluster Profiles

Cluster 1 consists of medium-sized firms with relatively high P/E ratios but lower profitability and growth rates. These firms appear to be moderately valued by investors despite weaker operating performance.

Cluster 2 contains the largest and most profitable firms. These firms have the highest market capitalization, strongest return on equity, and highest return on assets. They represent mature and financially strong pharmaceutical companies.

Cluster 3 contains the smallest firms in terms of market capitalization. These firms have the highest leverage and strongest revenue growth. This suggests they may be growth-oriented firms that rely more heavily on debt financing to support expansion.

```
cluster_summary <- aggregate(
  pharma_numeric,
  by = list(Cluster = pharma$Cluster),
  FUN = mean
)

cluster_summary
```

##	Cluster	Market_Cap	Beta	PE_Ratio	ROE	ROA	Asset_Turnover	Leverage
## 1	1	21.75500	0.5950000	46.90000	11.3	5.100000	0.7500000	0.3050000
## 2	2	97.11364	0.4336364	20.95455	35.7	14.954545	0.8000000	0.3254545
## 3	3	9.23500	0.6483333	19.43333	17.3	5.983333	0.4833333	1.2500000

##	Rev_Growth	Net_Profit_Margin
## 1	7.01000	6.65000
## 2	10.16455	20.17273
## 3	23.49000	13.51667

Cluster Visualization

The cluster plot shows clear separation among the three groups of pharmaceutical firms.

Cluster 2 is well separated from the other clusters and represents firms with stronger profitability and larger market capitalization.

Cluster 3 appears distinct because of its higher leverage and growth characteristics.

The visualization confirms that the k-Means algorithm successfully identified meaningful groups within the dataset.

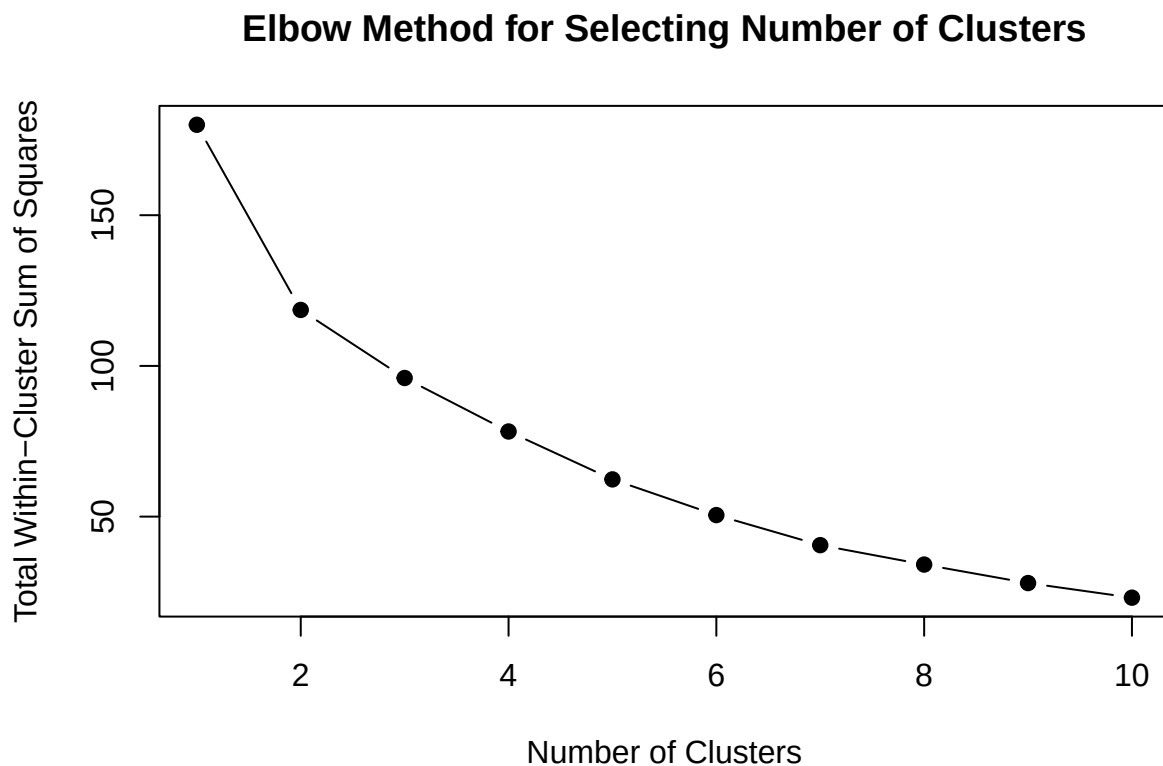
Determine Appropriate Number of Clusters

```
set.seed(123)

wss <- numeric(10)

for (k in 1:10) {
  kmeans_result <- kmeans(pharma_scaled, centers = k, nstart = 25)
  wss[k] <- kmeans_result$tot.withinss
}

plot(
  1:10,
  wss,
  type = "b",
  pch = 19,
  xlab = "Number of Clusters",
  ylab = "Total Within-Cluster Sum of Squares",
  main = "Elbow Method for Selecting Number of Clusters"
)
```



Apply k-Means Clustering

```
set.seed(123)

kmeans_model <- kmeans(
  pharma_scaled,
  centers = 3,
  nstart = 25
)

kmeans_model

## K-means clustering with 3 clusters of sizes 4, 11, 6
##
## Cluster means:
##   Market_Cap      Beta  PE_Ratio      ROE      ROA Asset_Turnover
## 1 -0.6125361  0.2698666  1.3143935 -0.9609057 -1.0174553    0.2306328
## 2  0.6733825 -0.3586419 -0.2763512  0.6565978  0.8344159    0.4612656
## 3 -0.8261772  0.4775991 -0.3696184 -0.5631589 -0.8514589   -0.9994088
##   Leverage Rev_Growth Net_Profit_Margin
## 1 -0.3592866 -0.5757385      -1.3784169
## 2 -0.3331068 -0.2902163       0.6823310
## 3  0.8502201  0.9158889      -0.3319956
##
## Clustering vector:
## [1] 2 1 1 2 3 1 2 3 3 2 2 3 2 3 2 2 2 1 2 3 2
##
## Within cluster sum of squares by cluster:
## [1] 20.54199 43.30886 32.14336
## (between_SS / total_SS =  46.7 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

Add Cluster Membership to Dataset

```
pharma$Cluster <- as.factor(kmeans_model$cluster)

pharma[, c("Symbol", "Name", "Cluster")]
```

```
##   Symbol      Name Cluster
## 1   ABT Abbott Laboratories    2
## 2   AGN   Allergan, Inc.     1
## 3   AHM   Amersham plc      1
## 4   AZN AstraZeneca PLC      2
## 5   AVE   Aventis          3
## 6   BAY   Bayer AG         1
```

## 7	BMJ	Bristol-Myers Squibb Company	2
## 8	CHT	Chattem, Inc	3
## 9	ELN	Elan Corporation, plc	3
## 10	LLY	Eli Lilly and Company	2
## 11	GSK	GlaxoSmithKline plc	2
## 12	IVX	IVAX Corporation	3
## 13	JNJ	Johnson & Johnson	2
## 14	MRX	Medicis Pharmaceutical Corporation	3
## 15	MRK	Merck & Co., Inc.	2
## 16	NVS	Novartis AG	2
## 17	PFE	Pfizer Inc	2
## 18	PHA	Pharmacia Corporation	1
## 19	SGP	Schering-Plough Corporation	2
## 20	WPI	Watson Pharmaceuticals, Inc.	3
## 21	WYE	Wyeth	2

Cluster Size

```
table(pharma$Cluster)
```

```
##
##  1  2  3
##  4 11  6
```

Cluster Centers

```
cluster_centers_scaled <- kmeans_model$centers
cluster_centers_scaled
```

##	Market_Cap	Beta	PE_Ratio	ROE	ROA	Asset_Turnover
## 1	-0.6125361	0.2698666	1.3143935	-0.9609057	-1.0174553	0.2306328
## 2	0.6733825	-0.3586419	-0.2763512	0.6565978	0.8344159	0.4612656
## 3	-0.8261772	0.4775991	-0.3696184	-0.5631589	-0.8514589	-0.9994088
##	Leverage	Rev_Growth	Net_Profit_Margin			
## 1	-0.3592866	-0.5757385	-1.3784169			
## 2	-0.3331068	-0.2902163	0.6823310			
## 3	0.8502201	0.9158889	-0.3319956			

Mean Financial Characteristics by Cluster

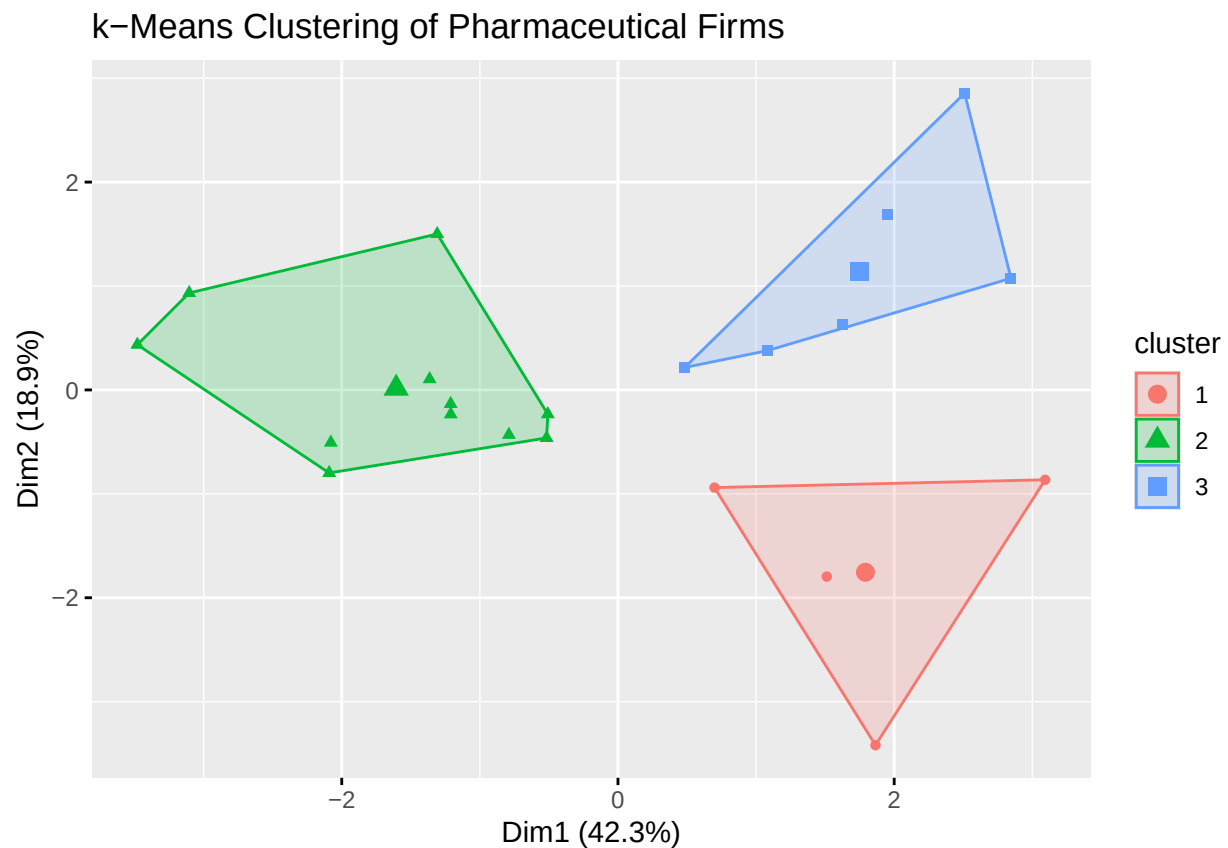
```
cluster_summary <- aggregate(
  pharma_numeric,
  by = list(Cluster = pharma$Cluster),
  FUN = mean
)
```

```
cluster_summary
```

```
##   Cluster Market_Cap      Beta PE_Ratio  ROE      ROA Asset_Turnover  Leverage
## 1      1    21.75500 0.5950000 46.90000 11.3  5.100000    0.7500000 0.3050000
## 2      2    97.11364 0.4336364 20.95455 35.7 14.954545    0.8000000 0.3254545
## 3      3     9.23500 0.6483333 19.43333 17.3  5.983333    0.4833333 1.2500000
##   Rev_Growth Net_Profit_Margin
## 1     7.01000      6.65000
## 2    10.16455     20.17273
## 3    23.49000     13.51667
```

Visualize the Clusters

```
fviz_cluster(  
  kmeans_model,  
  data = pharma_scaled,  
  geom = "point",  
  ellipse.type = "convex",  
  main = "k-Means Clustering of Pharmaceutical Firms"  
)
```



Interpret Clusters Using Non-Clustering Variables

```
table(pharma$Cluster, pharma$Median_Recommendation)
```

```
##
##      Hold Moderate Buy Moderate Sell Strong Buy
##  1      2          1          0          1
##  2      6          3          2          0
##  3      1          3          2          0
```

```
table(pharma$Cluster, pharma$Location)
```

```
##
##      CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
##  1          1      0          1      0          0 1 1
##  2          0      0          0      0          1 2 8
##  3          0      1          0      1          0 0 4
```

```
table(pharma$Cluster, pharma$Exchange)
```

```
##
##      AMEX NASDAQ NYSE
##  1      0      0      4
##  2      0      0     11
##  3      1      1      4
```

Cluster Profiles

Based on the cluster summary, the three clusters represent distinct groups of pharmaceutical firms.

Cluster 1 contains 4 firms with lower market capitalization, lower ROE, lower ROA, and the lowest net profit margin. This group also has the highest average P/E ratio, suggesting that investors may still value these firms despite weaker current profitability.

Cluster 2 contains 11 firms and represents the strongest financial group. This cluster has the highest average market capitalization, highest ROE, highest ROA, highest asset turnover, and highest net profit margin. These firms appear to be mature, financially strong pharmaceutical companies.

Cluster 3 contains 6 firms with the lowest market capitalization, highest beta, highest leverage, and strongest revenue growth. These firms appear to be smaller, growth-oriented firms that rely more heavily on debt financing.

Discussion and Conclusion

This assignment used k-Means clustering to group pharmaceutical firms based on nine numerical financial variables. The variables were standardized before clustering because they were measured on different scales. For example, market capitalization is measured in billions of dollars, while beta and leverage are measured as ratios.

The elbow method was used to support the selection of the number of clusters. A three-cluster solution was selected because it provides a reasonable balance between simplicity and meaningful segmentation.

The resulting clusters summarize how pharmaceutical firms differ in terms of size, risk, profitability, asset use, leverage, growth, and profit margin. The non-clustering variables, including analyst recommendation, headquarters location, and exchange, were used to examine whether the financial clusters also show meaningful business patterns.

Discussion

The clustering analysis demonstrates that pharmaceutical firms differ substantially in their financial characteristics. Market capitalization, profitability, leverage, and growth measures contribute to the formation of distinct clusters.

The results suggest the presence of three major business profiles:

1. Mature, financially strong firms.
2. Moderate-sized firms with average performance.
3. Smaller, growth-oriented firms with higher leverage.

These findings could help investors, analysts, and managers better understand competitive positioning within the pharmaceutical industry. Overall, k-Means clustering provides a useful exploratory machine learning approach for identifying groups of firms with similar financial profiles.

Conclusion

This assignment applied the k-Means clustering algorithm to a dataset of 21 pharmaceutical companies using nine financial variables.

All variables were standardized before clustering to ensure comparability across different measurement scales. The elbow method suggested that three clusters provided an appropriate balance between simplicity and explanatory power.

The resulting clusters revealed meaningful differences among firms based on size, profitability, leverage, and growth characteristics. Cluster 2 contained the largest and most profitable firms, Cluster 1 represented moderately sized firms with average performance, and Cluster 3 contained smaller firms with higher leverage and stronger growth rates.

Overall, the analysis demonstrates that k-Means clustering is an effective exploratory machine learning technique for identifying groups of firms with similar financial profiles. ““